Quantitative Research Methods

Multiple Testing

Advanced module A7 — optional self-study

Prof. Dr. Christoph Weisser

The advanced modules at a glance

Module	Topic
A1	Randomised Controlled Trials — potential outcomes, randomisation, power
A2	Explainable AI with Shapley Values — axioms, exact and Monte-Carlo Shapley
A3	Conformal Prediction — split conformal, CQR, prediction sets
A4	GLMs and Splines — exponential family, overdispersion, P-splines
A5	Support Vector Machines — margins, the cost C, kernels
A6	Survival Analysis — censoring, Kaplan–Meier, Cox
A7	**Multiple Testing** — FWER, Bonferroni and Holm, FDR
A8	Unsupervised Learning — PCA, K-means, hierarchical clustering

Eight optional, self-study modules that sit outside the taught chapter plan; today's module is highlighted. Each is a full deck in the course house style with a companion lab notebook.

This module assumes Chapter 0 (hypothesis testing and p-values) and Chapter 5 (resampling). It was ISLP Chapter 13 in earlier editions of this course.

Contents

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this chapter

Symbol	Meaning
m	number of hypotheses tested simultaneously
α	per-test significance level; target FWER
H_{0j}	j-th null hypothesis in the family
p_j, $p_{(j)}$	p-value of test j; the j-th smallest (sorted) p-value
V	number of false positives (true nulls rejected)
R	total number of rejections, $R = V + S$
S	number of true positives (false nulls rejected)
U, T	true nulls retained; false nulls retained (missed effects)
FWER	family-wise error rate $\Pr(V \geq 1)$
FDR	false discovery rate $E\left[V/\max(R, 1)\right]$
q	target FDR level of Benjamini–Hochberg
Power	$E[S]/m_1$, with m_0 and m_1 the number of true and false nulls ($m_0 + m_1 = m$)

Sources and references

Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python.* Springer.

Learning objectives

By the end of this chapter you will be able to:

- **Explain** the multiple-testing problem and its consequences.
- **Apply** Bonferroni and Holm corrections to control family-wise error rate (FWER).
- **Apply** the Benjamini–Hochberg procedure to control false discovery rate (FDR).
- **Use** resampling-based p-values when the null distribution is hard to derive.
- **Avoid** the post-selection-inference and p-hacking pitfalls.

Roadmap of this chapter

Honest inference at scale

1. The multiple-testing problem.
2. FWER controls: Bonferroni, Holm.
3. FDR control: Benjamini–Hochberg.
4. Resampling-based *p*-values.
5. Practical issues: *p*-hacking, post-selection inference.

Outline

Why naive testing fails at scale

Test m hypotheses simultaneously, each at level $\alpha = 0.05$.

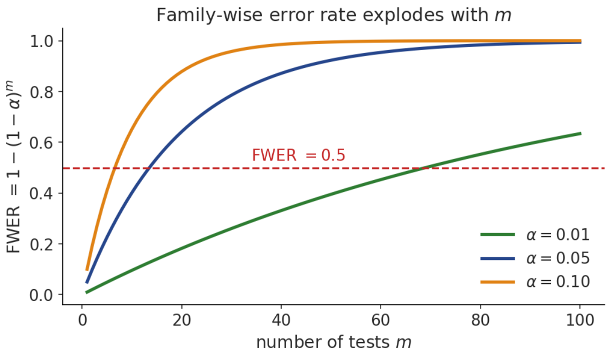
The arithmetic

- Even if all m nulls are true, we expect $0.05m$ false rejections.
- Test 10 000 genes $\Rightarrow \sim$ 500 false discoveries by chance.
- Naive thresholding fails dramatically.

Why “some metric always wins”

Score one experiment on dozens of metrics and segments and this arithmetic guarantees that *something* comes back significant — every time, whether or not the change does anything. That is the mechanism by which a null change gets shipped to every user.

Why the family-wise error rate explodes



Takeaway

With $\alpha = 0.05$ the chance of ≥ 1 false positive passes 50% by only $m = 14$ tests and approaches certainty as m grows.

The classical hypothesis-test refresher

How to read this

- H_0: null hypothesis. H_1: alternative.
- Type I error: reject H_0 when true.
- Type II error: fail to reject when false.
- p-value: probability under H_0 of a test statistic at least as extreme as observed.
- "Reject H_0 if $p < \alpha$" controls the per-test Type I error rate at α.

Takeaway

A p-value is computed *assuming* H_0 is true; it is **not** the probability that H_0 is true, and a small p is not proof of a large effect.

The decision-table view

	Not rejected	Rejected	Total
H_0 true	U	V	m_0
H_0 false	T	S	m_1
Total	$m - R$	R	m

How to read this

- Family-wise error rate (FWER): $\Pr(V \geq 1)$.
- False discovery rate (FDR): $E[V/R \mid R > 0] \Pr(R > 0)$.
- Power: $E[S]/m_1$.
- Read it as an alert queue and the letters become staffing: R is the alerts raised, V those that waste an investigator's day, V/R the false-positive rate operations argues about, and T the fraud nobody ever sees.

Exercise 13.1 — The multiple-testing problem [Concept]

Exercise

1. Define a Type I error and a Type II error in terms of H_0.
2. Suppose m hypotheses are tested independently, each at level $\alpha = 0.05$, and *all* nulls are in fact true. Write the probability of making at least one false rejection (the FWER) as a function of m.
3. Evaluate this probability for $m = 10$.
4. Find the smallest m for which the FWER reaches 0.5.

Solution — Exercise 13.1 (1/2)

Solution

Method. The FWER is easiest to reach through its complement: rather than count "at least one false positive" directly, find the single event "no false positive at all" and subtract from 1.

Part 1 — error types.

- Type I (false positive): rejecting H_0 when it is true.
- Type II (false negative): failing to reject H_0 when it is false.

Part 2 — FWER formula. One test avoids a false positive with probability $1-\alpha$. Under independence the m tests avoid it *jointly* with probability $(1-\alpha)^m$ (independent probabilities multiply), so the complement is

$$\mathrm{FWER} = \Pr(\text{at least one false positive}) = 1-(1-\alpha)^m.$$

Common mistake

Adding the per-test rates to get $m\alpha$ — that only approximates the FWER (the first-order term) and can exceed 1.

Solution — Exercise 13.1 (2/2)

Solution

Part 3 — $m = 10$.

$$1 - (0.95)^{10} = 1 - 0.5987 = 0.4013 \approx 40\%.$$

A 5% per-test rate has ballooned to a 40% chance of at least one spurious "discovery". **Part 4 — FWER $= 0.5$.** Solve $1 - 0.95^m \geq 0.5$, i.e. $0.95^m \leq 0.5$:

$$m \geq \frac{\ln 0.5}{\ln 0.95} = \frac{-0.6931}{-0.05129} = 13.51,$$

so $m = 14$ tests already give a better-than-even chance of a false positive. This is exactly why corrections are needed.

Common mistake

Dividing by $\ln 0.95$ without flipping the inequality. Since $\ln 0.95 < 0$, $0.95^m \leq 0.5 \iff m \ln 0.95 \leq \ln 0.5 \iff m \geq \ln 0.5 / \ln 0.95$ — the sign flips at the last step.

Outline

Bonferroni: split the error budget equally

Rule

$$\boxed{\text{Reject } H_{0j} \text{ if } p_j \le \alpha/m.}$$

How to read this

- m: number of tests in the family; α: target FWER (e.g. 0.05).
- Divide the budget evenly: each test is judged at the stricter level α/m.
- Equivalently, adjust each p-value to $\tilde{p}_j = \min(1, m\, p_j)$ and compare to α.

Takeaway

Controls FWER at α under *any* dependence. Conservative: it throws away power, but is simple, popular, and a good starting point.

Holm's step-down procedure (uniformly beats Bonferroni)

Algorithm

1. Order p-values $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$.
2. Find the smallest j with $p_{(j)} > \alpha/(m-j+1)$.
3. Reject all $H_{0(k)}$ for $k < j$.

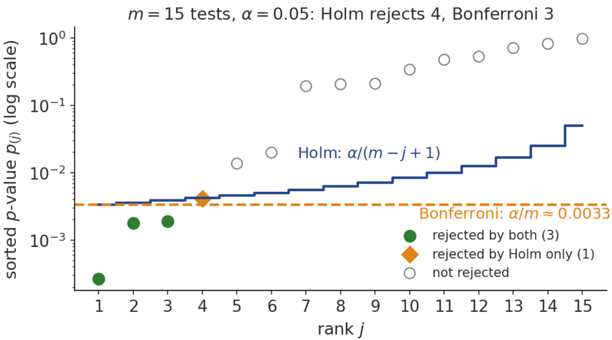
Why it is more powerful

- The thresholds *grow* down the list: $\frac{\alpha}{m}, \frac{\alpha}{m-1}, \ldots, \alpha$ — the first equals Bonferroni, every later one is looser.
- "Step-down": start at the smallest p-value and walk down, stopping at the first test that fails; keep all larger p-values.

Takeaway

Uniformly more powerful than Bonferroni, yet still controls FWER at α.

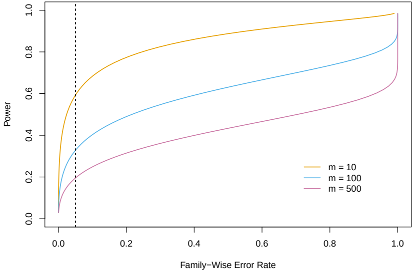
Bonferroni vs. Holm on 15 simulated p-values



Takeaway

Bonferroni's flat bar $\alpha/m \approx 0.0033$ rejects 3 hypotheses; Holm's rising thresholds $\alpha/(m - j + 1)$ additionally catch $p_{(4)} = 0.0041$, rejecting 4 — Holm always rejects at least as many as Bonferroni.

Power loss with FWER control



Takeaway

Power collapses as m grows: at FWER $= 0.05$ (dashed) it falls from about 0.6 at $m = 10$ to about 0.2 at $m = 500$ — FWER control costs real effects.

Source: Figure 13.5 — ISLP, James et al. (2023).

Exercise 13.2 — Bonferroni correction [Math]

Exercise

You run $m = 6$ tests and obtain the sorted p-values

$$p = (0.001,\ 0.008,\ 0.012,\ 0.030,\ 0.040,\ 0.600),$$

and wish to control the FWER at $\alpha = 0.05$.

1. State the Bonferroni rejection threshold.
2. Which hypotheses are rejected?
3. Compute the Bonferroni-adjusted p-values $\tilde{p}_j = \min(1,\ m\, p_j)$ and confirm they give the same decisions.

Solution — Exercise 13.2

Solution

Step 1 — threshold. $\alpha/m = 0.05/6 = 0.00833$.

Step 2 — reject if $p_j \le 0.00833$.

p_j	0.001	0.008	0.012	0.030	0.040	0.600
≤ 0.00833?	✓	✓	×	×	×	×

Reject H_1 and H_2 only ($0.008 < 0.00833$; $0.012 > 0.00833$).

Step 3 — adjusted p-values $\tilde{p}_j = \min(1, 6p_j)$:

$$(0.006,\ 0.048,\ 0.072,\ 0.180,\ 0.240,\ 1.000).$$

Rejecting when $\tilde{p}_j \le 0.05$ again flags exactly H_1 (0.006) and H_2 (0.048). Two rejections.

Take-away. Bonferroni is simple and valid under any dependence, but conservative — here it misses H_3 at $p = 0.012$.

Common mistake: correcting twice — compare raw p_j with α/m, *or* adjusted $\tilde{p}_j$ with α. Comparing $\tilde{p}_j$ with α/m double-penalises.

Exercise 13.3 — Holm's step-down procedure [Math]

Exercise

Using the same data as Exercise 13.2,

$$p = (0.001,\ 0.008,\ 0.012,\ 0.030,\ 0.040,\ 0.600), \quad m = 6,\ \alpha = 0.05,$$

apply Holm's step-down procedure. For each rank j compare $p_{(j)}$ with $\alpha/(m-j+1)$, stop at the first failure, and report the rejections. Compare with Bonferroni.

Solution — Exercise 13.3

Solution

The p-values are already sorted. Compare each with its Holm threshold $\alpha/(m-j+1)$:

j	$p_{(j)}$	$\alpha/(m-j+1)$	decision
1	0.001	$0.05/6 = 0.00833$	$0.001 \leq 0.00833$ ✓
2	0.008	$0.05/5 = 0.01000$	$0.008 \leq 0.01000$ ✓
3	0.012	$0.05/4 = 0.01250$	$0.012 \leq 0.01250$ ✓
4	0.030	$0.05/3 = 0.01667$	$0.030 > 0.01667$ $\times$

The first failure is at $j = 4$. Holm rejects all hypotheses with rank < 4, i.e. $H_{(1)}, H_{(2)}, H_{(3)}$ — **three** rejections.

Comparison. Holm also rejects $H_{(3)}$ at $p = 0.012$, which Bonferroni missed. Holm is uniformly more powerful than Bonferroni while still controlling the FWER at α, because its first threshold matches Bonferroni's but every later one is larger.

Common mistake: continuing down the list after the first failure. Holm is *step-down*: once $p_{(4)}$ fails, ranks 4–6 are all retained — even if a later $p_{(j)}$ were to sit below its own threshold.

Outline

The Benjamini–Hochberg (BH) procedure

Algorithm (control FDR at level q)

1. Order the p-values: $p_{(1)} \leq \cdots \leq p_{(m)}$.
2. Find the largest L such that $p_{(L)} \leq Lq/m$.
3. Reject the L smallest hypotheses.

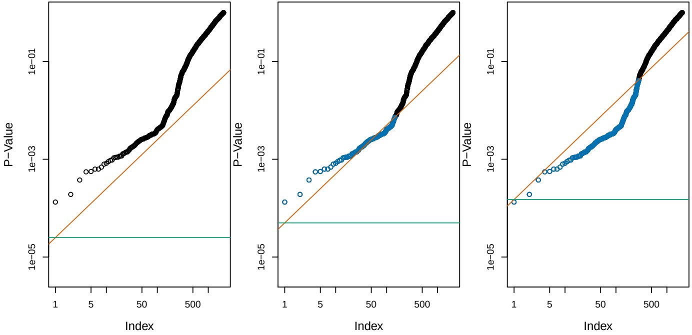
Reading the threshold Lq/m

- q: target FDR (e.g. 0.10); m: number of tests; L: rank in the sorted list.
- The bar Lq/m *rises* with rank, unlike Bonferroni's flat α/m.
- "Step-up": scan from the *largest* p-value down; reject everything up to the first one that clears its bar.

Takeaway

Under independence (or positive dependence), BH controls FDR at q.

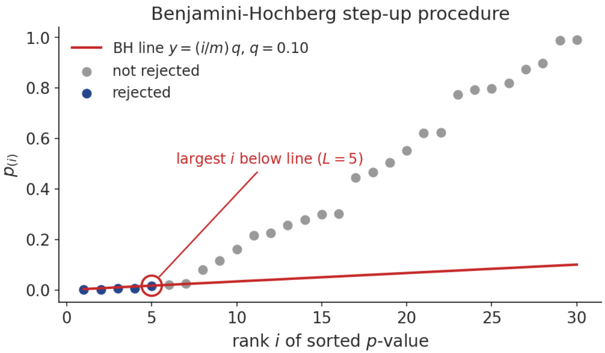
Visualising BH



Sorted p-values vs. the line Lq/m. Reject everything below the highest crossing.

Source: Figure 13.6 — ISLP, James et al. (2023).

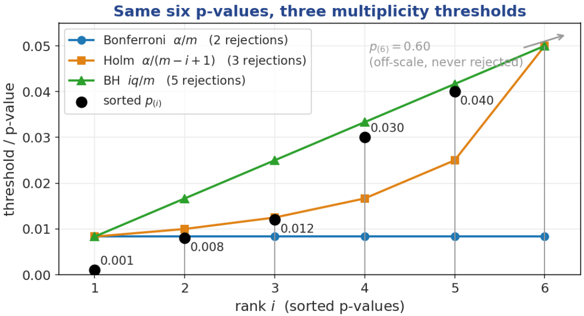
Benjamini–Hochberg as a staircase



Takeaway

Compare each sorted $p_{(i)}$ with the line $(i/m)q$; take the *largest* rank below it and reject everything up to there (5 of 30 here at $q = 0.10$).

Three thresholds on the same six p-values



Takeaway

A p-value is a candidate for rejection when it sits *below* its method's curve. All three meet at rank 1; then Bonferroni stays flat while Holm and BH rise — so here they reject 2, 3 and 5 hypotheses (Exercises 13.2/13.3/13.5).

FDR vs. FWER: which to use?

How to read this

- Use **FWER** when even one false discovery is unacceptable (drug-approval trials).
- Use **FDR** when an exploratory list of candidates is acceptable and false positives are tolerable in moderation (genomics, A/B testing).
- BH is much more powerful than Bonferroni when m is large and a non-trivial fraction of nulls are false.

Ask what the output is for

A false claim that is expensive and *public* — a drug label, a filing, a change shipped to every user — calls for FWER. A *ranked shortlist for further investigation* calls for FDR, because a known fraction of false leads is an acceptable cost of doing business. Where it is regulated the choice is not yours: FDA guidance on multiple endpoints requires a *pre-specified* multiplicity strategy.

Exercise 13.4 — FDR vs. FWER [Concept]

Exercise

1. Give the definitions of the FWER and the FDR in terms of the decision-table quantities V (false rejections) and R (total rejections).
2. Explain why controlling the FDR is generally less conservative (more powerful) than controlling the FWER.
3. For each scenario, state whether FWER or FDR control is more appropriate and why: (i) a confirmatory phase-III drug trial; (ii) screening 20 000 genes for follow-up experiments.

Solution — Exercise 13.4 (1/2)

Solution

Part 1 — definitions.

$$\text{FWER} = \Pr(V \geq 1), \qquad \text{FDR} = E\left[\frac{V}{R} \,\middle|\, R > 0\right] \Pr(R > 0).$$

FWER is the probability of *any* false positive; FDR is the *expected proportion* of false positives among the rejections.

Part 2 — why FDR is more powerful. FWER guards against a single false positive anywhere, so its threshold must shrink with m (roughly α/m). FDR only requires the false positives to be a small *fraction* of discoveries, so it tolerates a few errors when there are many true discoveries. This lets it use a larger, adaptive threshold and reject far more when many nulls are false.

Solution — Exercise 13.4 (2/2)

Solution

Part 3 — scenarios.

(i) **FWER** (Bonferroni/Holm): a single false approval is unacceptable; the cost of any Type I error is very high.

(ii) **FDR** (Benjamini–Hochberg): an exploratory candidate list is the goal; a modest fraction of false leads is acceptable and will be filtered by follow-up experiments. FWER here would be far too conservative and kill power.

Common mistake

Defaulting to FWER "to be safe". Safety is not free — the stricter criterion buys its guarantee with missed true effects, so the choice is a design decision, not a formality.

Exercise 13.5 — Benjamini–Hochberg procedure [Math]

Exercise

Using the same six p-values,

$$p = (0.001,\ 0.008,\ 0.012,\ 0.030,\ 0.040,\ 0.600), \quad m = 6,$$

apply the Benjamini–Hochberg (BH) procedure at FDR level $q = 0.05$.

1. For each rank L compute the BH threshold Lq/m.
2. Find the largest L with $p_{(L)} \leq Lq/m$.
3. State the rejections and compare with Bonferroni and Holm.

Solution — Exercise 13.5 (1/2)

Solution

Steps 1–2 — compare each $p_{(L)}$ with Lq/m, where $q/m = 0.05/6 = 0.00833$:

L	$p_{(L)}$	Lq/m	$p_{(L)} \leq Lq/m$?
1	0.001	0.00833	✓
2	0.008	0.01667	✓
3	0.012	0.02500	✓
4	0.030	0.03333	✓
5	0.040	0.04167	✓
6	0.600	0.05000	×

The largest L satisfying the condition is $L = 5$.

Solution — Exercise 13.5 (2/2)

Solution

Step 3 — reject. BH rejects the 5 smallest hypotheses $H_{(1)}, \ldots, H_{(5)}$ (everything except $p = 0.600$).

Comparison on identical data.

$$\text{Bonferroni: } 2 \quad < \quad \text{Holm: } 3 \quad < \quad \text{BH: } 5 \text{ rejections.}$$

Controlling the FDR yields many more discoveries than controlling the FWER — at the price of allowing a small expected fraction ($\leq q$) of them to be false. Note BH is a *step-up* rule: even though $p_{(4)} = 0.030$ and $p_{(5)} = 0.040$ individually clear their thresholds here, BH would reject them via the largest passing L regardless of any earlier borderline case.

Common mistake

Treating q like α. They bound different things: α caps the chance of *any* false rejection; q caps only the expected *fraction* of the rejected list that is false.

Extended Exercise 13.1 — Full workflow on ten p-values [Math]

Extended exercise (15 min)

Ten hypotheses give the sorted p-values

$$p = (0.0008, 0.0041, 0.0060, 0.0100, 0.0180, 0.0290, 0.0850, 0.150, 0.240, 0.790),$$

with $m = 10$. Control the FWER at $\alpha = 0.05$ (Bonferroni, Holm) and the FDR at $q = 0.10$ (Benjamini–Hochberg).

1. List the rejections under **Bonferroni**, **Holm** and **BH**.
2. Verify the ordering $R_{\text{Bonf}} \leq R_{\text{Holm}} \leq R_{\text{BH}}$.
3. State the guaranteed FWER bound, and comment on what the FDR guarantee buys.

Solution — Extended Exercise 13.1 (1/3)

Solution — Bonferroni and Holm (FWER ≤ 0.05)

Bonferroni: reject if $p_j \leq \alpha/m = 0.005$. Only $p_{(1)} = 0.0008$ and $p_{(2)} = 0.0041$ qualify $\Rightarrow$ **2 rejections**.
Holm: compare $p_{(j)}$ with $\alpha/(m-j+1)$, stopping at the first failure:

j	$p_{(j)}$	$\alpha/(m-j+1)$	
1	0.0008	$0.05/10 = 0.00500$	✓
2	0.0041	$0.05/9 = 0.00556$	✓
3	0.0060	$0.05/8 = 0.00625$	✓
4	0.0100	$0.05/7 = 0.00714$	× stop

Holm rejects ranks 1–3 $\Rightarrow$ **3 rejections**.

Solution — Extended Exercise 13.1 (2/3)

Solution — Benjamini–Hochberg (FDR ≤ 0.10)

Compare each $p_{(L)}$ with $Lq/m = 0.01L$ and take the *largest* passing L:

L	$p_{(L)}$	Lq/m	
5	0.0180	0.050	✓
6	0.0290	0.060	✓ ← largest
7	0.0850	0.070	×
8	0.150	0.080	×

(Ranks 1–4 pass too.) The largest L with $p_{(L)} \leq Lq/m$ is $L = 6$, so BH rejects the six smallest $\Rightarrow$ **6 rejections** — including $p_{(6)} = 0.029$ and ranks $4, 5$ that the FWER rules missed.

Solution — Extended Exercise 13.1 (3/3)

Solution — comparison and error guarantees

$$\underbrace{2}_{\text{Bonferroni}} \leq \underbrace{3}_{\text{Holm}} \leq \underbrace{6}_{\text{BH}}.$$

FWER bound. Bonferroni and Holm both guarantee $\mathrm{FWER} = \Pr(V \geq 1) \leq \alpha = 0.05$ under *any* dependence: the chance of *even one* false rejection is at most 5%. **What FDR buys.** BH instead guarantees

$\mathrm{FDR} = E[V/R] \leq q\, m_0/m \leq 0.10$. Among its 6 discoveries, at most an *expected* 10% ($\lesssim 1$ here) are false. Tolerating a small false *fraction* rather than *any* false positive is exactly why BH rejects more.

Common mistake

Reading "FDR $\leq$ 0.10" as "each discovery is 90% likely true". The FDR is an average over the whole rejected list (and over repetitions), not a per-discovery probability.

Outline

p-hacking and selection bias

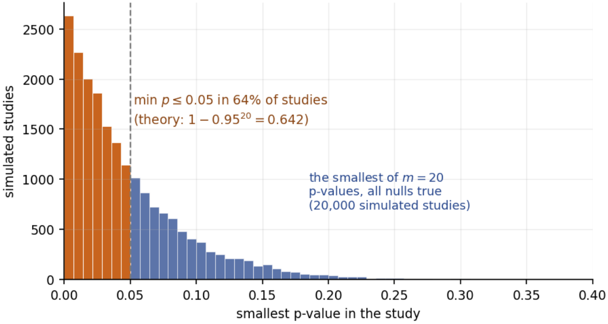
Don't

- Report only the most-significant subset of tests.
- Test thousands of hypotheses and call the smallest p-value "significant".

Takeaway

Multiple-testing correction must include *every* test you ran, not only the ones you report. Pre-register hypotheses where possible; out-of-sample replication is the gold standard. Backtesting hundreds of strategy variants on one price history is the same failure wearing a different hat: an impressive Sharpe ratio arrives by chance, which is why practitioners deflate reported performance for the number of trials and demand out-of-time evidence before allocating capital.

What p-hacking harvests, drawn



Takeaway

Test $m = 20$ outcomes on pure noise and report the best: 64% of studies hand you a "significant" result ($1 - 0.95^{20} = 0.642$ — the same arithmetic as the FWER explosion, now read as a publication strategy). The smallest p-value in a fishing expedition is not a p-value.

Outline

Python lab — run it live in the notebook

Companion notebook: `advanced_07_multiple_testing_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- §1, *Simulation* — $m = 1000$ tests with a fraction of true alternatives;
- §2, *Compare procedures* — Bonferroni, Holm and Benjamini–Hochberg side by side;
- §3, *Visualise the Benjamini–Hochberg line*, then §4, *Permutation test*.

Data loads via the `ISLP` package or the bundled CSV files; §5, *Exercises* ends the notebook with hands-on tasks.

Bonferroni, Holm, BH in statsmodels

```python
from statsmodels.stats.multitest import multipletests # corrections
import numpy as np # arrays

p = np.array([1e-6, 0.01, 0.04, 0.05, 0.30, 0.45, 0.80]) # m = 7 tests

for method in ["bonferroni", "holm", "fdr_bh"]: # FWER, FWER, FDR control
    rej, p_corr, *_ = multipletests(p, alpha=0.05, method=method) # adjust
    print(f"{method:12s}", rej) # True = reject H_j
    print(f"{'p_adj':12s}", np.round(p_corr, 4)) # compare with 0.05
```

How to read this

- **In:** the raw p-values of the *whole* family ($m = 7$ here) and the level $\alpha = 0.05$; `fdr_bh` reads it as the FDR target q.
- **Out:** `rej`, the reject flags, and `p_corr`, the *adjusted* p-values — pre-scaled (Bonferroni: $\min(1, mp_j)$) so you compare them with α itself, never with α/m.
- The printouts rank as the chapter predicts: Bonferroni fewest, Holm at least as many, BH most.

Extended Exercise 13.3 — Simulating FDR and power [Python]

Extended exercise (15 min)

Simulate $m = 1000$ one-sided z-tests of which $m_1 = 100$ have a true effect (mean shift $\mu = 3$); the other 900 are null. Over many repeats, apply Benjamini–Hochberg at $q = 0.10$ and Bonferroni at $\alpha = 0.05$, and estimate the *realised* FDR (V/R) and power (S/m_1) of each.

1. Write the simulation and run ~ 2000 repeats.
2. Report the realised FDR and power for BH and for Bonferroni.
3. Check BH against its theoretical bound $q\, m_0/m$, and comment on the power gap.

Run this live

The worked solution sits in `advanced_07_multiple_testing_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise 13.3 — Simulating FDR and power [Python]*.

Solution — Extended Exercise 13.3 (1/3)

Solution — code: setup and the BH rule

```python
import numpy as np # arrays and random numbers
from scipy import stats # normal cdf for p-values
rng = np.random.default_rng(0) # seeded generator: reproducible
m, m1, mu = 1000, 100, 3.0 # 100 true effects, mean shift mu
q, alpha = 0.10, 0.05 # FDR target q, FWER target alpha
truth = np.arange(m) < m1 # ground truth: V and S need it

def bh_reject(p, q): # Benjamini--Hochberg step-up
    order = np.argsort(p); ps = p[order] # rank 1 = smallest p
    thr = (np.arange(1, m + 1) / m) * q # rising bar: (j/m) * q
    hit = np.where(ps <= thr)[0] # every rank under its bar
    rej = np.zeros(m, bool) # start: reject nothing
    if len(hit): rej[order[:hit.max() + 1]] = True # step up to largest
    return rej # mask in original order
```

Solution — Extended Exercise 13.3 (2/3)

Solution — code: simulate FDR and power

```
def simulate(method, reps=2000): # average FDR / power over reps
    fdr = pw = 0.0 # running totals
    for _ in range(reps): # one repeat = one study
        z = rng.normal(0, 1, m); z[:m1] = rng.normal(mu, 1, m1) # effects: mean mu
        p = 1 - stats.norm.cdf(z) # one-sided p-values
        rej = bh_reject(p, q) if method == "bh" else (p <= alpha / m) # BH or Bonf.
        V = (rej & ~truth).sum(); S = (rej & truth).sum(); R = rej.sum() # decision table
        fdr += V / max(R, 1); pw += S / m1 # FDP of this study; max avoids 0/0
    return fdr / reps, pw / reps # FDR = mean FDP over studies

for meth in ["bh", "bonferroni"]: # compare both procedures
    f, pw = simulate(meth) # run the experiment
    print("%-11s FDR=%.3f power=%.3f" % (meth, f, pw)) # read on the next slide
```

Solution — Extended Exercise 13.3 (3/3)

Solution — expected numbers and interpretation

Typical printout.

```
bh   FDR=0.090   power=0.723
bonferroni   FDR=0.002   power=0.186
```

- **BH** holds the realised FDR at 0.090 — essentially its theoretical ceiling $q\,m_0/m = 0.10 \times 900/1000 = 0.09$ — while recovering about **72%** of the true effects.
- **Bonferroni** is far stricter: realised FDR ≈ 0.002, but power collapses to about **19%** — it misses four-fifths of the signals.
- **Lesson:** at $m = 1000$, FDR control detects roughly *four times* as many true effects as FWER control, for a small, bounded false fraction.

Common mistake: expecting the realised FDR to hit $q = 0.10$ exactly. The guarantee is $q\,m_0/m = 0.09$ here — BH adapts to the unknown share of true nulls, and any single run fluctuates around it.

Outline

Chapter 13 in one slide

What we accomplished

Honest inference when many hypotheses are tested at once.

- Set up the multiple-testing problem.
- Family-wise error rate (FWER): Bonferroni and Holm.
- False discovery rate (FDR): Benjamini–Hochberg.
- Resampling-based p-values and Storey's q-values.
- Pitfalls of post-selection inference and p-hacking.

Key formulas at a glance

FWER $\mathrm{FWER} = \Pr(V \geq 1) = 1 - (1-\alpha)^m$ (independent tests).

Bonferroni reject H_{0j} if $p_j \leq \alpha/m$ (controls FWER).

Holm reject while $p_{(j)} \leq \dfrac{\alpha}{m-j+1}$, stepping down from the smallest.

FDR $\mathrm{FDR} = E\left[\dfrac{V}{\max(R,1)}\right]$ (expected false-discovery proportion).

Benjamini–Hochberg reject the L smallest, where $L = \max\{j : p_{(j)} \leq \frac{j}{m} q\}$.

BH guarantee $\mathrm{FDR} = E[V/R] \leq q\, m_0/m \leq q$ (m_0 true nulls; independence or positive dependence).

Power $\mathrm{Power} = E[S]/m_1$.

Vocabulary checklist

Term	**Meaning**
Type I error	Reject H_0 when true
Type II error	Fail to reject when false
FWER $= \Pr(V \geq 1)$	Family-wise error rate
FDR $= E[V/R]$	False discovery rate
Bonferroni / Holm	FWER controls
BH procedure	Step-up FDR control
Storey's q-value	FDR-calibrated p-value
Permutation test	Null distribution by relabelling
Knockoffs	Modern FDR control with covariate engineering

Decision rules of thumb

- Always specify the family of tests *before* looking at the data.
- Confirmatory / regulated $\Rightarrow$ FWER (Bonferroni or Holm).
- Exploratory / discovery $\Rightarrow$ BH on FDR.
- Report both raw and adjusted p-values.
- Replicate findings on an independent sample whenever possible.
- For valid post-selection inference: sample-splitting, selective inference, knockoffs.

Common pitfalls

Don't

1. Test thousands of hypotheses and report only the "significant" ones.
2. Use BH when tests are strongly negatively dependent.
3. Confuse FWER and FDR — they answer different questions.
4. Apply post-hoc multiple-testing fixes after looking at the data.
5. Use uncorrected p-values from stepwise / lasso.
6. Treat a p-value as a probability that H_0 is true.

Course-wide summary

The conceptual arc

1. Vocabulary and bias-variance (Ch. 1–2).
2. Linear baselines and how to evaluate them (Ch. 3–5).
3. Regularisation and dimension reduction (Ch. 6).
4. Adding flexibility: splines, GAMs, trees, SVMs, deep nets (Ch. 7–10).
5. Special responses: censored times, no response (Ch. 11–12).
6. Honest inference at scale (Ch. 13).

Closing thoughts

The four habits of an effective statistical learner

1. Always plot the data.
2. Always validate on data you didn't train on.
3. Always interpret what the model says about the world.
4. Always check assumptions and limitations before claiming results.

Thank you for taking this course. Good luck applying statistical learning to your own data!

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the chapter: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **Extended Exercise 13.2 — FWER or FDR? A design decision** — Exercise 13.4 expanded with arithmetic, over three solution slides. Run it from here if the concept version left doubts; otherwise it is homework.
- **The four outcomes, drawn** — a diagram of the same 2×2 table the main deck already gives as a table.
- **Resampling-based inference** — permutation p-values, the resampling estimate of the FDR, and what the histogram of many p-values looks like. Use it when the null distribution is not available in closed form.
- **Post-selection inference** — why a p-value computed after looking at the data is not a p-value. Important in research practice, beyond the exam.

Extended Exercise 13.2 — FWER or FDR? A design decision [Integrative]

Extended exercise (15 min)

For each study, choose an error criterion and justify it, using the naive expected number of false discoveries $E[V] = m_0\,\alpha$ (all-null worst case $m_0 = m$) and the power consequences.

1. **Scenario A — confirmatory phase-III trial:** $m = 4$ co-primary endpoints; one false "significant" endpoint could license an ineffective drug.
2. **Scenario B — exploratory screen:** $m = 20\,000$ genes tested for follow-up; a shortlist with a few false leads is acceptable.
3. Contrast the statistical power the two criteria leave on the table.

Solution — Extended Exercise 13.2 (1/3)

Solution — scenario A: control FWER

With $m = 4$ tests at $\alpha = 0.05$, naive testing expects

$$E[V] = m_0\alpha = 4(0.05) = 0.20 \text{ false positives,} \qquad \mathrm{FWER} = 1 - 0.95^4 = 0.185.$$

A $\sim$19% chance of *any* false endpoint is unacceptable when the decision is regulatory and irreversible. **Choose FWER control** (Bonferroni threshold $\alpha/m = 0.0125$, or Holm). One false positive is the outcome we must not allow, so we pay the power cost gladly — with only $m = 4$ tests that cost is tiny.

Solution — Extended Exercise 13.2 (2/3)

Solution — scenario B: control FDR

With $m = 20\,000$ and (worst case) $m_0 \approx 20\,000$, naive testing expects

$$E[V] = m_0\alpha = 20\,000(0.05) = 1000 \text{ false discoveries!}$$

Bonferroni would shrink the threshold to $\alpha/m = 2.5 \times 10^{-6}$, cutting $E[V]$ to ≈ 0.05 but **destroying power** for realistic effect sizes. **Choose FDR control** (BH at $q = 0.10$): accept that an expected $\leq 10\%$ of the shortlist is false in exchange for detecting far more true signals — the false leads are removed by follow-up experiments.

Solution — Extended Exercise 13.2 (3/3)

Solution — the power trade-off

	Confirmatory ($m = 4$)	Exploratory ($m = 20\,000$)
Criterion	FWER (Bonferroni/Holm)	FDR (BH)
Threshold shape	α/m (fixed, tiny)	adaptive Lq/m
Naive $E[V]$	0.20	1000
Priority	no false positive	many true findings

Trade-off. The FWER threshold shrinks like α/m, so power collapses as m grows — fine for a handful of endpoints, ruinous for 20 000 genes. FDR asks only that false positives be a small *fraction* of discoveries, so its adaptive threshold keeps power high at scale. **Rule of thumb:** match the criterion to the cost of a single false positive.

Common mistake

Quoting $E[V] = m\alpha$ as a fact about your study — it is the all-null worst case ($m_0 = m$). With real effects present the count is smaller, though still ruinous at scale.

The four outcomes of a test decision

	Not rejected	**Rejected**	
H_0 true	U true negatives	V Type I (false pos.)	m_0
H_0 false	T Type II (false neg.)	S true positives	m_1
	$m - R$	$R = V + S$	

Takeaway

Type I errors are cell V — rejected true nulls. The FWER $= \Pr(V \geq 1)$ guards against *any* false positive; the FDR $= E[V/R]$ controls their *expected fraction* among the R rejections.

Resampling-based p-values

When the test statistic's null distribution is hard to derive:

How to read this

- **Permutation**: shuffle the response many times to get a null distribution.
- **Bootstrap (under H_0)**: resample under a constraint that enforces the null.
- Empirical p-values are then thresholded by Bonferroni / Holm / BH as before.

Takeaway

Empirical p-value $= \dfrac{1 + \#\{\text{resampled statistics} \geq \text{observed}\}}{1 + B}$ over B resamples — no parametric null needed. The "$+1$" keeps it strictly positive.

Resampling-based FDR

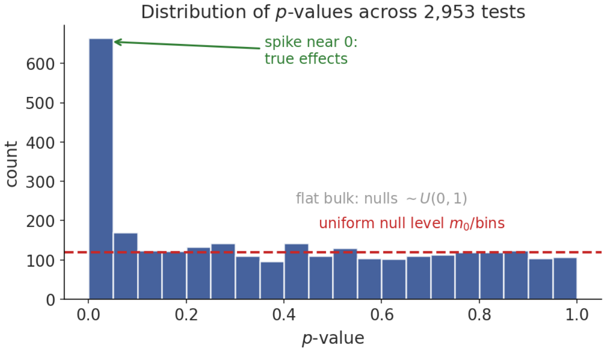
Storey's q-value

Estimate the proportion of true nulls $\hat{\pi}_0$ from the p-value distribution. Calibrate FDR more accurately than BH. Especially relevant when many nulls are false.

How $\hat{\pi}_0$ is read off

True nulls give p-values that are $\sim U(0,1)$, so the histogram is *flat* away from 0; the height of that flat part estimates $\hat{\pi}_0$. BH implicitly assumes $\hat{\pi}_0 = 1$; plugging in the smaller true value makes FDR control less conservative and recovers more discoveries.

The distribution of p-values under many tests



Takeaway

True nulls give p-values that are $\sim U(0,1)$ (the flat bulk), while true effects pile up near 0. Storey's method reads $\hat{\pi}_0$ off the height of that flat part to calibrate the FDR.

Post-selection inference

After variable selection (lasso, stepwise) standard p-values are invalid.

Takeaway

The bias: the *same data* both *chose* the hypothesis and *tested* it, so the reported p-values are optimistically small. Valid inference must account for the selection step.

Modern methods

- Sample splitting / data-carving.
- Selective inference (Lee, Sun, Sun, Taylor).
- Knockoffs (Barber, Candès).